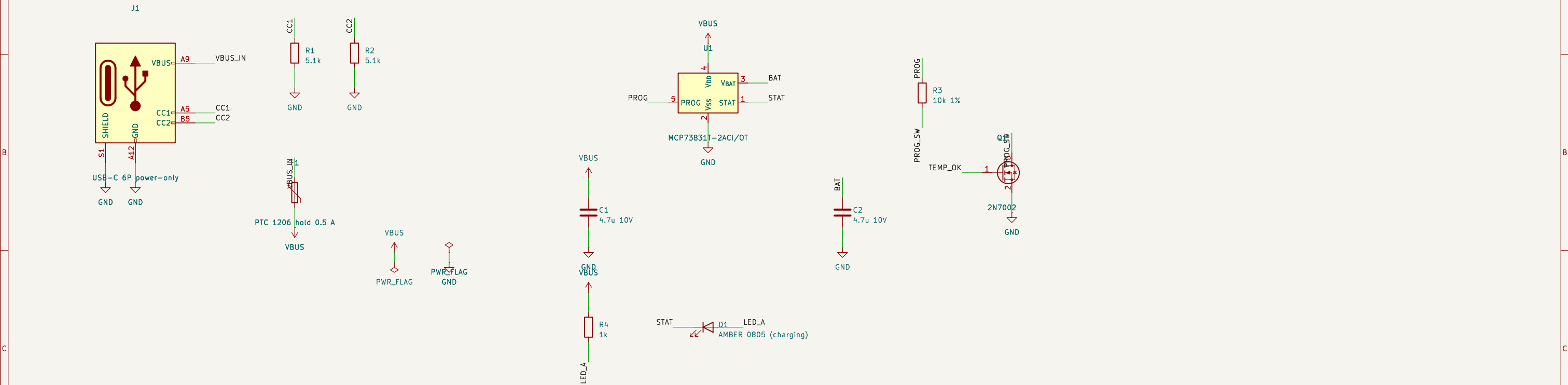


1	2	3	4	5	6	7	8
NIVA CHARGING DOCK – ENGINEERING PROTOTYPE – NOT FOR FABRICATION							

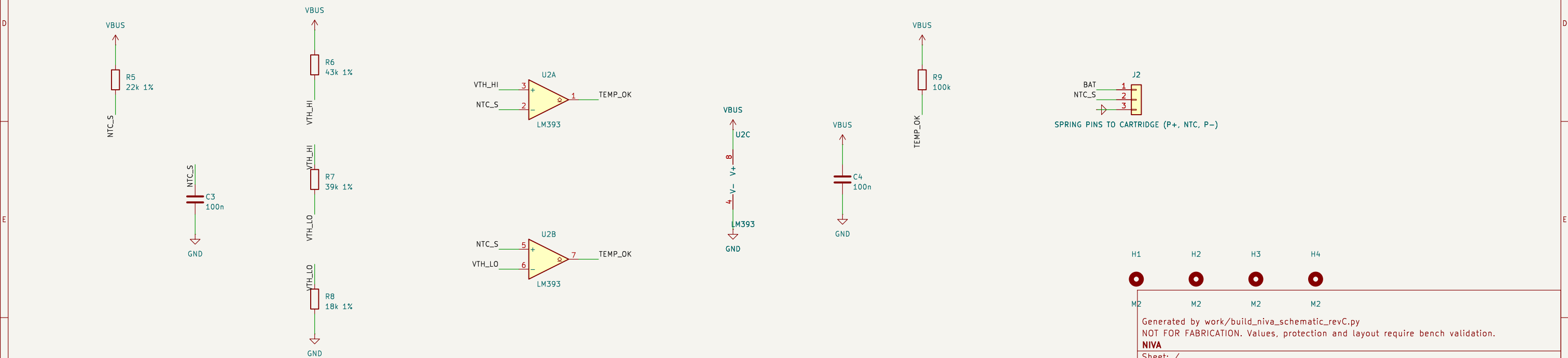
01 USB-C SINK (5 V ONLY)

02 LINEAR CHARGER 4.20 V / 100 mA



R_{PROG} 10k -> I_{REG} = 1000 V / 10 kOhm = 100 mA (MCP73831 datasheet).
 PROG reaches GND only through Q1: Q1 off -> PROG floats -> charger disabled.
 D1 lights while STAT is low (charging). Charge complete or inhibited = off.

03 PACK NTC WINDOW (HARDWARE, NO FIRMWARE)



NTC_S = $VBUS \times R_{NTC} / (22k + R_{NTC})$. Window (ratio of VBUS): cold/open trip 0.57 (VTH_HI), hot/short trip 0.18 (VTH_LO).
 10k B3435 NTC: 0 C $\rightarrow$ 28.7k $\rightarrow$ 0.566; 45 C $\rightarrow$ 4.85k $\rightarrow$ 0.181. B3950: trips at -2.5 C / -43 C (conservative).
 No cartridge: NTC_S = VBUS (5 V) $\rightarrow$ unit A pulls TEMP_OK low $\rightarrow$ no charge. Only VTH_HI (2.85 V) must be inside the
 LM393 common-mode range (VCC - 1.5 V at 25 C, VCC - 2 V over temperature); an input outside it still gives a correct output.
 No hysteresis: charge may toggle at a threshold, which is benign (charging is inhibited outside the window).

File: NIVA-dock.kicad_sch

Title: NIVA cartridge charging dock | USB-C, MCP73831, hardware NT

Size: A3	Date: 2026-09-28	Rev: C - ENGINEER
KiCad E.D.A. 9.0.9	Id: 1/1	